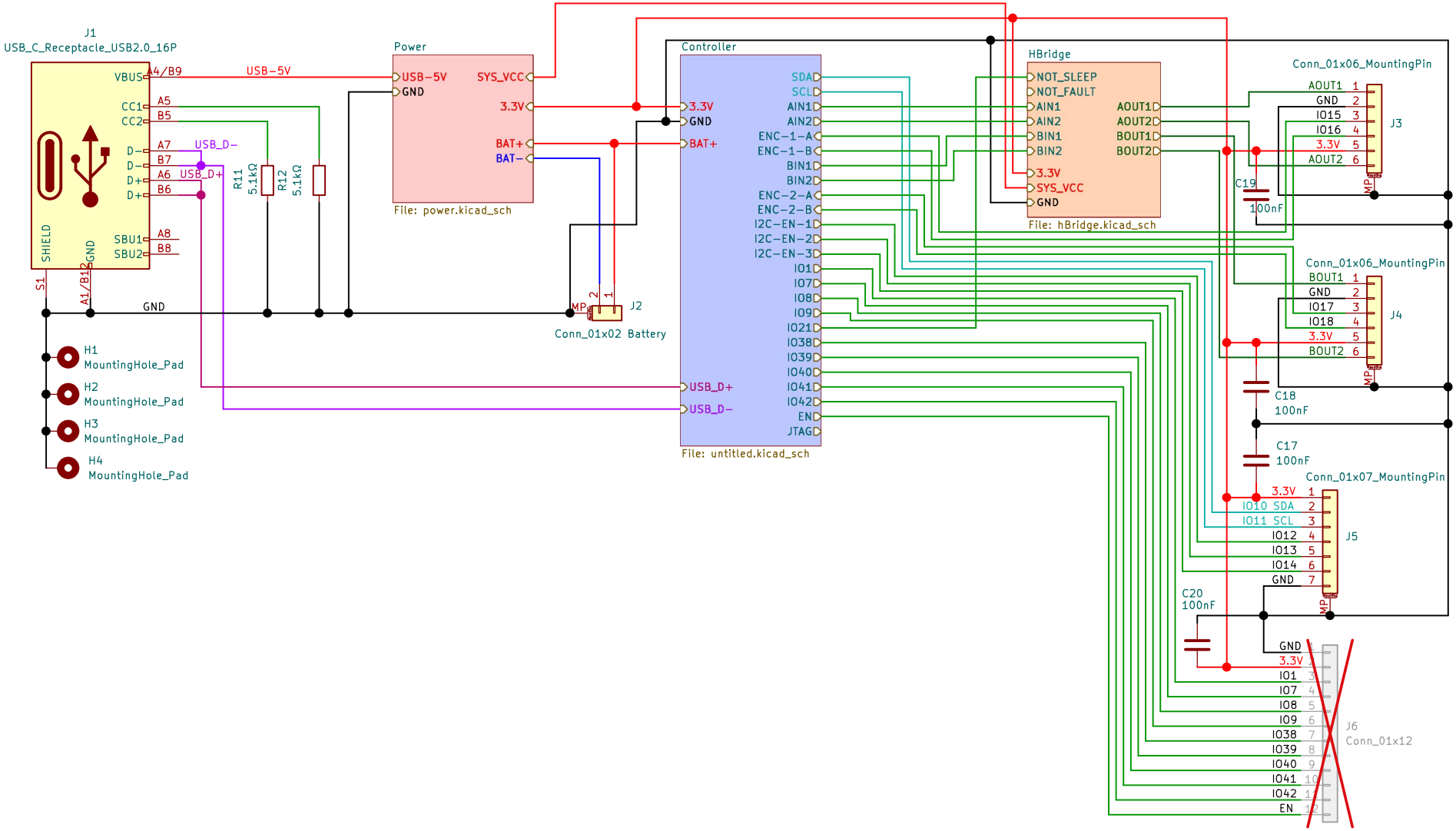
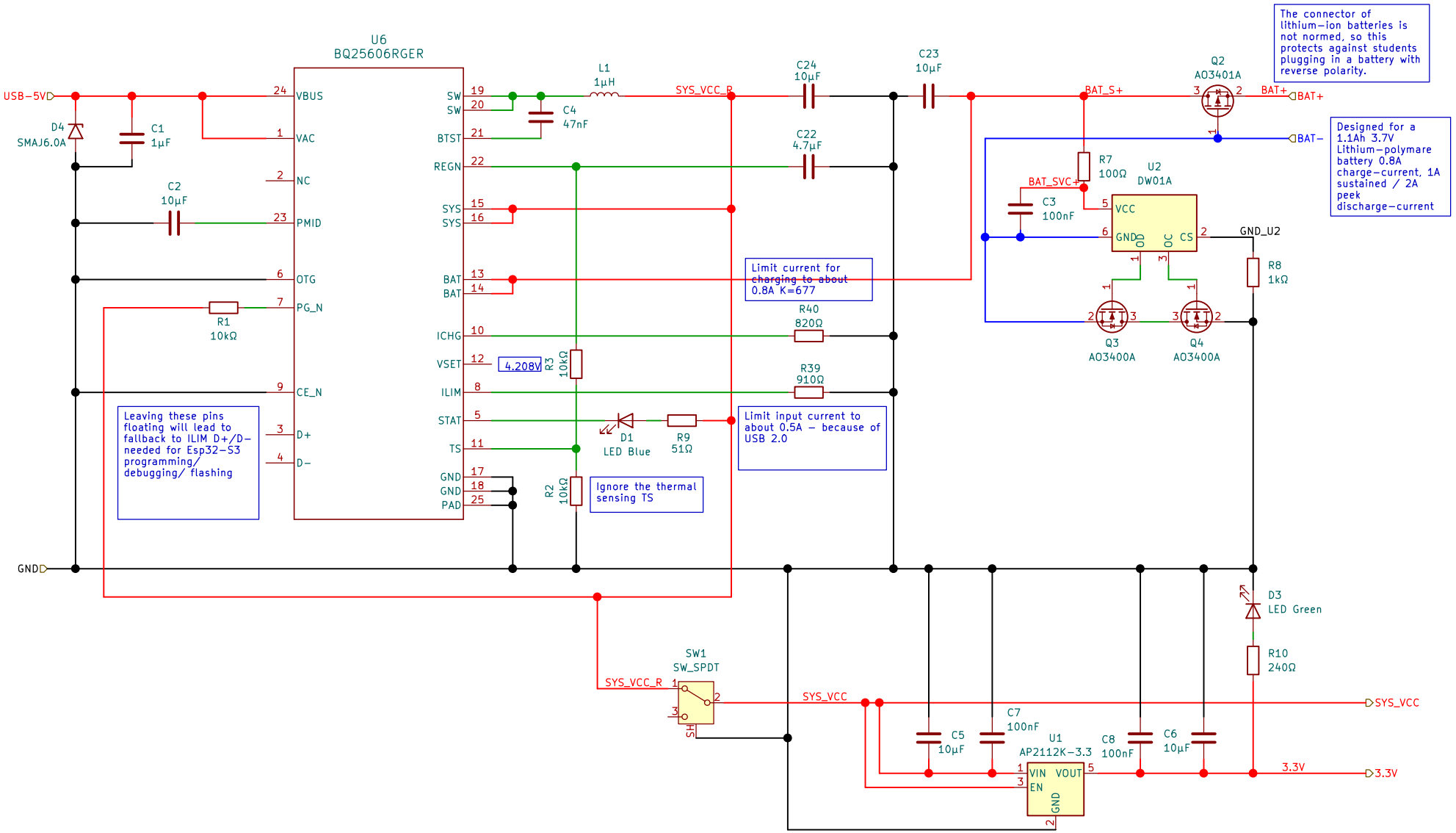
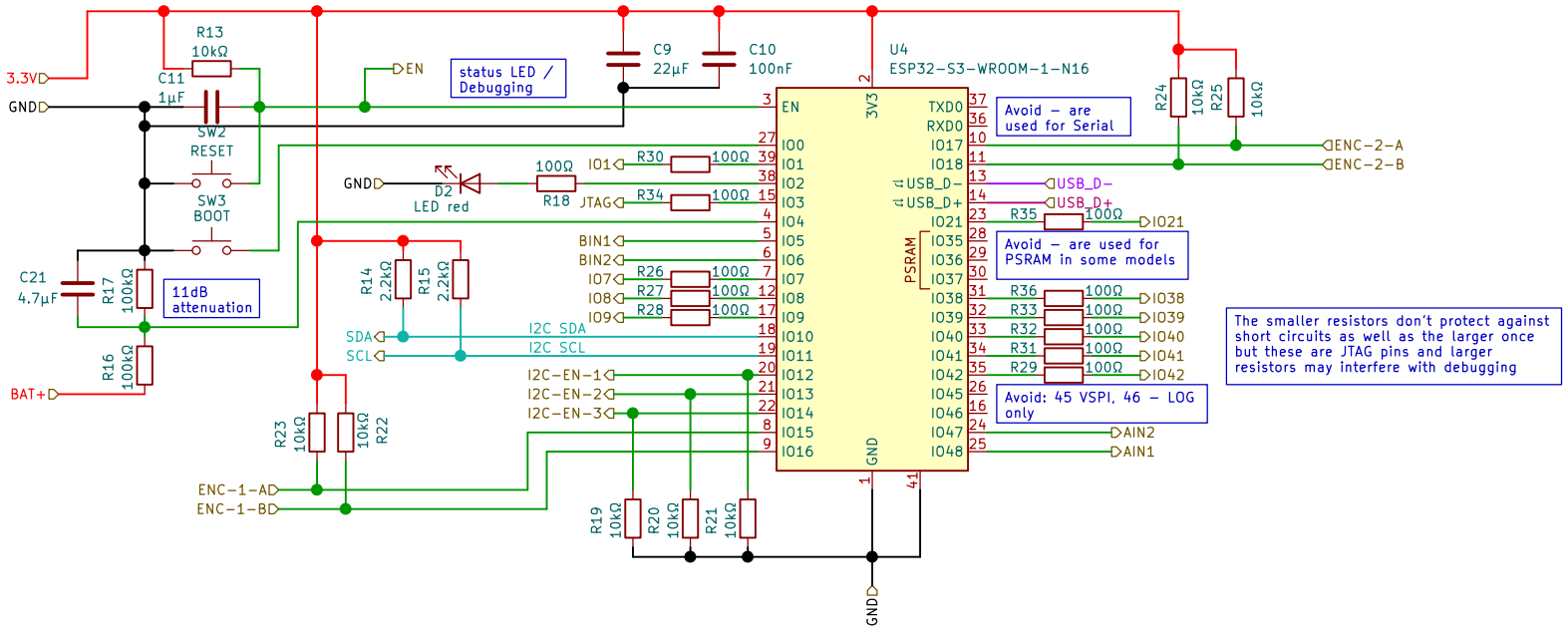
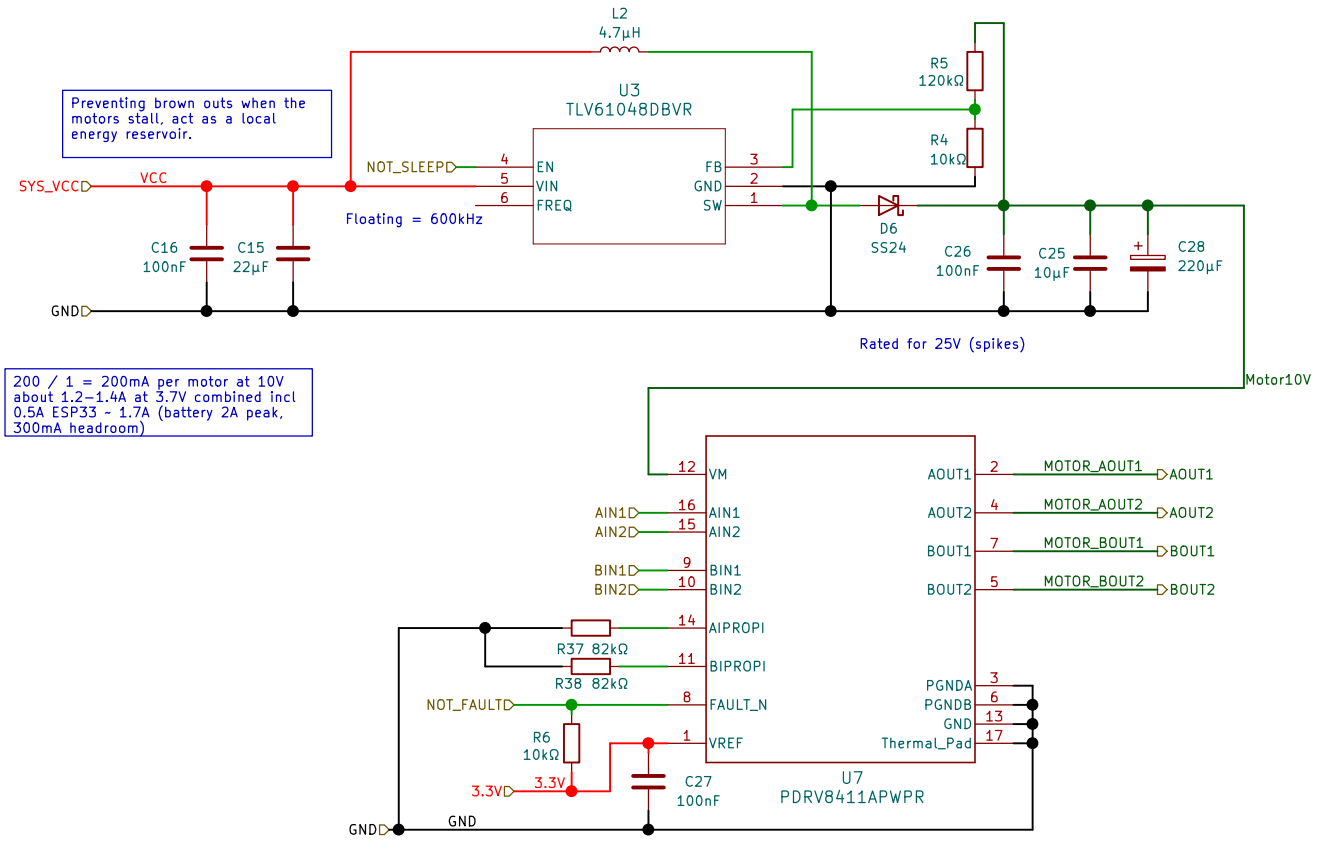

J1
USB_C_Receptacle_USB2.0_16P
VBUS
A4/B9
USB-5V
CC1
CC2
A5
B5
D-
D-
D+
D+
A7
B7
A6
B6
USB_D-
USB_D+
R11
5.1kΩ
R12
5.1kΩ
SHIELD
GND
SBU1
SBU2
A8
B8
S1
A1/B12
GND
H1
MountingHole_Pad
H2
MountingHole_Pad
H3
MountingHole_Pad
H4
MountingHole_Pad
Power
USB-5V
GND
SYS_VCC
3.3V
BAT+
BAT-
File: power.kicad_sch
MP
2
1
J2
Conn_01x02 Battery
Controller
3.3V
GND
BAT+
USB_D+
USB_D-
SDA
SCL
AIN1
AIN2
ENC-1-A
ENC-1-B
BIN1
BIN2
ENC-2-A
ENC-2-B
I2C-EN-1
I2C-EN-2
I2C-EN-3
IO1
IO7
IO8
IO9
IO21
IO38
IO39
IO40
IO41
IO42
EN
JTAG
File: untitled.kicad_sch
HBridge
NOT_SLEEP
NOT_FAULT
AIN1
AIN2
BIN1
BIN2
3.3V
SYS_VCC
GND
AOUT1
AOUT2
BOUT1
BOUT2
File: hBridge.kicad_sch
Conn_01x06_MountingPin
AOUT1 1
GND 2
IO15 3
IO16 4
3.3V 5
AOUT2 6
J3
MP
C19
100nF
Conn_01x06_MountingPin
BOUT1 1
GND 2
IO17 3
IO18 4
3.3V 5
BOUT2 6
J4
MP
C18
100nF
C17
100nF
Conn_01x07_MountingPin
3.3V 1
IO10 SDA 2
IO11 SCL 3
IO12 4
IO13 5
IO14 6
GND 7
J5
MP
C20
100nF
GND
3.3V
IO1
IO7
IO8
IO9
IO38
IO39
IO40
IO41
IO42
EN
J6
Conn_01x12


U6
BQ25606RGER
USB-5V
D4
SMAJ6.0A
C1
1µF
C2
10µF
R1
10kΩ
VBUS
VAC
NC
PMID
OTG
PG_N
CE_N
D+
D-
SW
BTST
REGN
SYS
BAT
ICHG
VSET
ILIM
STAT
TS
GND
PAD
L1
1µH
C4
47nF
R3
10kΩ
4.208V
R2
10kΩ
Ignore the thermal sensing TS
D1
LED Blue
R9
51Ω
SYS_VCC_R
C24
10µF
C23
10µF
C22
4.7µF
R40
820Ω
R39
910Ω
Limit current for charging to about 0.8A K=677
Limit input current to about 0.5A – because of USB 2.0
Leaving these pins floating will lead to fallback to ILIM D+/D- needed for Esp32-S3 programming/ debugging/ flashing
Q2
AO3401A
BAT_S+
BAT+
BAT-
The connector of lithium-ion batteries is not normed, so this protects against students plugging in a battery with reverse polarity.
Designed for a 1.1Ah 3.7V Lithium-polymare battery 0.8A charge-current, 1A sustained / 2A peek discharge-current
R7
100Ω
BAT_SVC+
C3
100nF
U2
DW01A
VCC
GND
OD
OC
CS
GND_U2
R8
1kΩ
Q3
AO3400A
Q4
AO3400A
GND
SW1
SW_SPDT
SYS_VCC_R
SYS_VCC
C5
10µF
C7
100nF
U1
AP2112K-3.3
VIN
VOUT
EN
GND
C8
100nF
C6
10µF
3.3V
D3
LED Green
R10
240Ω


3.3V
GND
R13
10kΩ
C11
1µF
SW2
RESET
SW3
BOOT
C21
4.7µF
R17
100kΩ
11dB
attenuation
R16
100kΩ
BAT+
EN
status LED /
Debugging
C9
22µF
C10
100nF
U4
ESP32-S3-WROOM-1-N16
GND
D2
LED red
100Ω
R18
IO1
R30
100Ω
JTAG
R34
100Ω
R14
2.2kΩ
R15
2.2kΩ
BIN1
BIN2
IO7
IO8
IO9
R26
R27
R28
100Ω
SDA
SCL
I2C SDA
I2C SCL
I2C-EN-1
I2C-EN-2
I2C-EN-3
R23
10kΩ
R22
10kΩ
ENC-1-A
ENC-1-B
R19
10kΩ
R20
10kΩ
R21
10kΩ
EN
3V3
GND
IO0
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
IO9
IO10
IO11
IO12
IO13
IO14
IO15
IO16
TXD0
RXD0
IO17
IO18
USB_D-
USB_D+
IO21
IO35
IO36
IO37
IO38
IO39
IO40
IO41
IO42
IO45
IO46
IO47
IO48
PSRAM
Avoid – are
used for Serial
Avoid – are used for
PSRAM in some models
Avoid: 45 VSPI, 46 – LOG
only
R24
10kΩ
R25
10kΩ
ENC-2-A
ENC-2-B
USB_D-
USB_D+
R35
100Ω
IO21
R36
R33
R32
R31
R29
100Ω
IO38
IO39
IO40
IO41
IO42
AIN2
AIN1
The smaller resistors don't protect against
short circuits as well as the larger once
but these are JTAG pins and larger
resistors may interfere with debugging


L2
4.7µH
U3
TLV61048DBVR
Preventing brown outs when the motors stall, act as a local energy reservoir.
NOT_SLEEPD
EN
VIN
FREQ
FB
GND
SW
SYS_VCC
VCC
Floating = 600kHz
R5
120kΩ
R4
10kΩ
D6
SS24
C16
100nF
C15
22µF
C26
100nF
C25
10µF
C28
220µF
GND
Rated for 25V (spikes)
200 / 1 = 200mA per motor at 10V about 1.2–1.4A at 3.7V combined incl 0.5A ESP33 ~ 1.7A (battery 2A peak, 300mA headroom)
Motor10V
VM
AIN1
AIN2
BIN1
BIN2
AIPROPI
BIPROPI
FAULT_N
VREF
AOUT1
AOUT2
BOUT1
BOUT2
MOTOR_AOUT1
MOTOR_AOUT2
MOTOR_BOUT1
MOTOR_BOUT2
PGNDA
PGNDB
GND
Thermal_Pad
R37 82kΩ
R38 82kΩ
NOT_FAULTD
R6
10kΩ
3.3VD
3.3V
C27
100nF
U7
PDRV8411APWPR
GNDD
GND